

# A High-resolution Image-based Virtual Try-on System in Taobao E-commerce Scenario

Zhilong Zhou

zhouzhilong.zzl@alibaba-inc.com

Alibaba Group  
Beijing, China

Tiezheng Ge

tiezheng.gtz@alibaba-inc.com

Alibaba Group  
Beijing, China

Shiyao Wang

shiyao.wsy@alibaba-inc.com

Alibaba Group  
Beijing, China

Yuning Jiang

mengzhu.jyn@alibaba-inc.com

Alibaba Group  
Beijing, China

## ABSTRACT

On an e-commerce platform, virtual try-on not only improves consumers' shopping experience but also attracts more consumers. However, the existing public virtual try-on dataset can't be generalized to Taobao fashion items because of low resolution, racial differences, and small dataset size. In this work, we build a large-scale and high-resolution virtual try-on dataset for Taobao e-commerce. Based on the dataset, we propose a virtual try-on system to generate high-resolution and attractive virtual try-on images without time-consuming image preprocessing. The system has covered more than 10000 fashion items and produced millions of virtual try-on results in Taobao e-commerce scenario.

## CCS CONCEPTS

• Information systems → Multimedia content creation.

## KEYWORDS

high-resolution virtual try-on system, e-commerce scenario

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## 1 INTRODUCTION

In Taobao, a world-leading e-commerce platform, clothing is one of the most important categories and covers major clothing brands. For a fashion item in Taobao, the model images always draw more attention than clothes images, which means more clicks and benefits for e-commerce platforms. The photo-realistic and high-resolution virtual try-on images are usually more attractive especially when the fashion items don't have model images. Image-based virtual

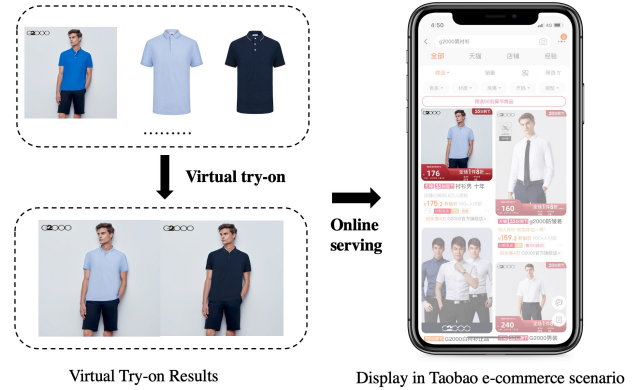


Figure 1: The pipeline of virtual try-on image generation and online display in Taobao.

try-on aims to generate a new model image  $I_m$  given a model image  $I'_m$  and a target clothes image  $I_c$ . Most existing virtual try-on methods[7][11][9][13][6][12] heavily depend on lots of image preprocessing results including human parsing[8], pose estimation[3], and densepose[5]. However, on the one hand, complicated image preprocessing is time-consuming and computationally inefficient during online serving. On the other hand, the photo-realistic virtual try-on results rely on high-quality image preprocessing results during training try-on models. Therefore we apply knowledge distillation mechanism in try-on models following [4], which avoids such costly preprocessing.

The training dataset plays an important role in virtual try-on methods. However, the existing virtual try-on datasets such as VITON[7], VITON-HD[7], and MPV[2] have some limitations in Taobao e-commerce scenario: low-resolution images (256x192), small dataset size (14221 image pairs), and racial differences (Caucasian). Therefore, in this work, we introduce a try-on data collection pipeline and build a large-scale and high-resolution virtual try-on dataset in Taobao e-commerce scenario. The dataset consists of 125838 image pairs ( $I_m, I_c$ ) with 800x800 resolution. Based on the dataset, we build a high-resolution virtual try-on system in the Taobao platform, as shown in Fig 1.

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## 2 DATASET COLLECTION

The collected dataset consists of  $(I_m, I_c) \in \mathcal{D}$ , in which  $I_c$  and  $I_m$  denote the target clothes image and model image where the model wears target clothes respectively. We first collect all fashion item images in Taobao e-commerce platform as candidate image set  $\mathcal{I}$ . Then face detection<sup>1</sup> algorithm module is employed to filter the candidate images  $\mathcal{I}$  and generate candidate image pairs set  $(I_m, I_c)$ , in which  $I_m$  and  $I_c$  represent images that have and not have detected faces respectively for the same fashion item. After face detection, there are two major problems: 1) clothes mismatch problem:  $(I_m, I_c)$  have different target clothes; 2) model/no model misclassification due to limitation of face detection. To tackle the misclassification problem, we annotate 1000 training images to train a logistic regression classifier to filter the images. For the pairs mismatch problem, we find that the color of clothes is a key factor for judging true positive image pairs. Therefore we employ the center cropping strategy, structural similarity, and RGB histogram similarity to discover the correct match of  $I_m$  and  $I_c$ . The image pairs with similarity scores below a given threshold  $t$  will be removed. It's worth noting that the above data cleaning strategies are also used in online serving, which makes it possible to automatically collect try-on image pairs every day. Fig 2 shows the dataset collection pipeline. The high-resolution virtual try-on dataset  $(I_m, I_c) \in \mathcal{D}$  contains 125838 image pairs with 800×800 resolution.

## 3 OUR SYSTEM

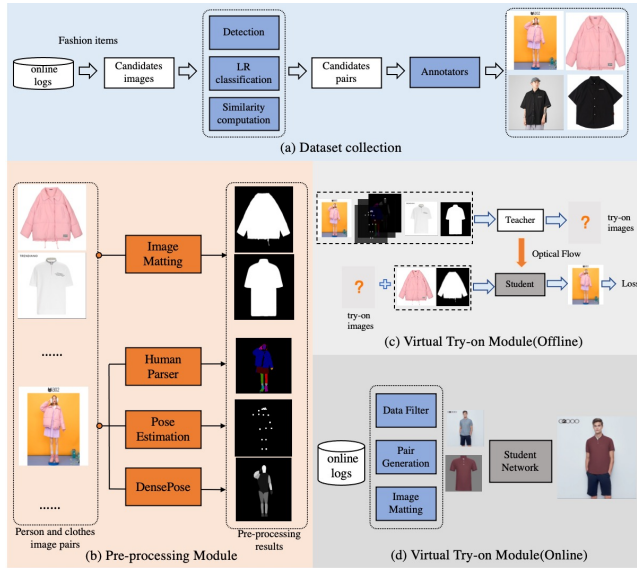


Figure 2: The pipeline of our system

### 3.1 Offline Training

The offline training of virtual try-on models consists of preprocessing and virtual try-on modules. The preprocessing module consists of human parsing[8], pose estimation[3], densepose[5] estimation,

<sup>1</sup><https://github.com/davisking/dlib>

and image matting as shown in Fig 2(b).  $I_m$  and  $I_c$  are preprocessed to obtain auxiliary information, which is sent to the following virtual try-on module.

The input of try-on module is image triplet  $(I_m, I_c, I'_c)$  and preprocessing results for  $I_m$ , in which  $I_m$  denotes model image where model wears  $I_c$ ,  $I_c$  and  $I'_c$  denote two different clothes images. The module is motivated by [4], which includes teacher network  $\mathcal{T}$  and student network  $\mathcal{S}$ , is shown in Fig 2(c). In each network, we warp the target clothes  $I_c$  by estimating appearance flow and then generate try-on images by Res-UNet[10] architecture. We have tried to estimate a thin plate spline transformation[1] to warp the clothes but it will distort the warped clothes when it comes to high resolution. The appearance flow can build a dense correspondence between the  $I_m$  and  $I_c$ , therefore we employ appearance flow[14] to produce high-resolution warped clothes. Training the try-on module consists of pre-training stage and jointly training stage. In the pre-training stage,  $\mathcal{T}$  is trained by image pairs  $(I_m, I_c)$  and preprocessing results as in [7]. In the joint training stage,  $I_m$ ,  $I'_c$  and preprocessing results are sent to teacher networks  $\mathcal{T}$  to generate try-on results  $I'_m$ . Generated  $I'_m$ , input  $I_c$  and clothes mask  $\mathcal{M}$  are sent to  $\mathcal{S}$  to generate try-on results  $I_m$ , which is shown in Figure 3. The estimated appearance flows in  $\mathcal{T}$  are more accurate because the preprocessing inputs carry richer semantic information. Therefore we distill the appearance flow between  $\mathcal{T}$  and  $\mathcal{S}$ , which transfers semantic information from  $\mathcal{T}$  and facilitates  $\mathcal{S}$  to get a more accurate estimation. Through joint training,  $\mathcal{S}$  can obtain human pose and location information from  $\mathcal{T}$  without preprocessing.

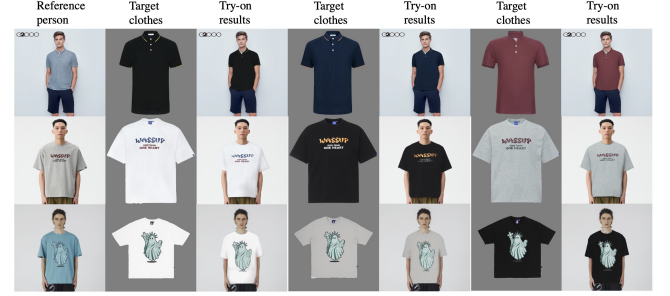


Figure 3: Virtual try-on visual results generated by our system on collected high-resolution dataset.

### 3.2 Online Serving

The student network  $\mathcal{S}$  has been deployed as the online virtual try-on module on Taobao e-commerce platform. All try-on image pairs are generated following the strategies described in Sec.2.  $\mathcal{S}$  receives image pairs to obtain try-on results, which are stored in the online materials database. The high-resolution virtual try-on system has covered more than 10,000 products and generated millions of try-on images on Taobao e-commerce platform. During online A/B testing on Taobao e-commerce platform, the generated try-on images contribute up to 4% CTR(Click Through Rate) promotion compared with the clothes images for the same fashion item. The improvement demonstrates that try-on images have a more attractive visual display than clothes images.

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